

Simplest Generative Model: Naïve Bayes

- Represent each sample in a feature space $x_i \in \text{VALUES}^d$, where *VALUES* is a finite set of possible feature values
- Assume the label is discrete $y \in L$, where L is some finite set
- **Model:** the Naïve Bayes model is defined as

$$P(x, y) = P(y) \prod_{j=1}^d P(x^{(j)} | y)$$

- That is, features are assumed to be independent given the label

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- **Prediction:** for an example $x_i \in R^d$

$$y^* = \operatorname{argmax}_y P(y)P(x|y)$$

- As there are only so many values y can take, we just iterate over all of them and find the maximum

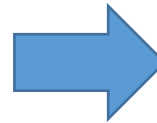
Example of a Naïve Bayes: Text Classification Task with Bag of Words Features

- Task: given text, predict what category it is from
- We assume that the topic is the label y , and represent the text as a count vector of the words in it
 - Each distinct wordform that may appear in the document is a feature
 - Values of features are usually indicators (0 or 1)

yossi likes apples

dina likes oranges

dana likes both apples and oranges



[1,0,0,1,1,0,0,0]

[0,1,0,1,0,1,0,0]

[0,0,1,1,1,1,1,1]

Features: *yossi, dina, dina, likes, apples, oranges, and, both*

- This works OK for text categorization if the topics are not too fine-grained

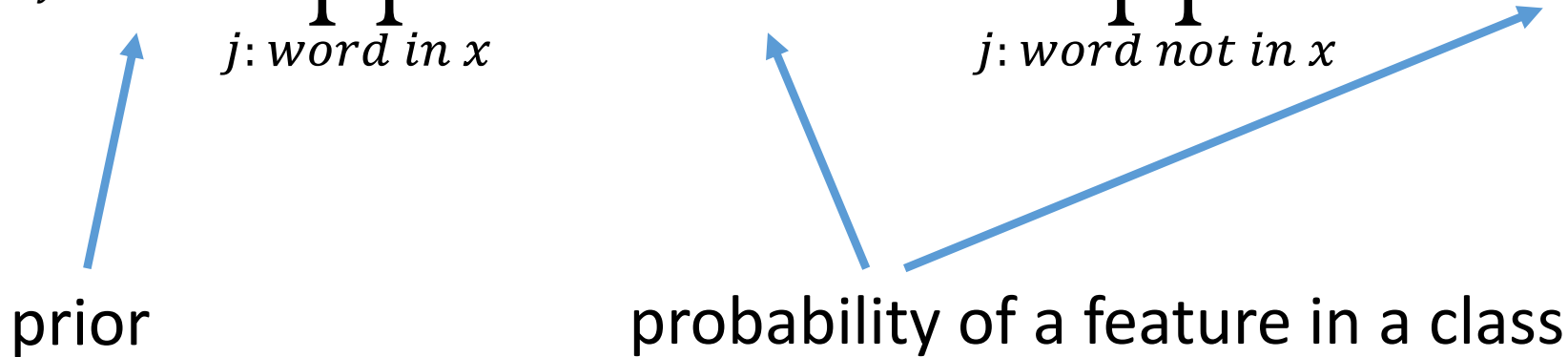
Prediction in Bag of Words Naïve Bayes

- Given a trained model, and a document x , we may predict the most likely y :

$$y^* = \operatorname{argmax}_y P(y)P(x|y) = \operatorname{argmax}_y P(y) \prod_{j=1}^d P(x^{(j)}|y) =$$

$$= \operatorname{argmax}_y P(y) \prod_{j: \text{word in } x} P(x^{(\text{word})} = 1|y) \prod_{j: \text{word not in } x} P(x^{(\text{word})} = 0|y)$$

prior



probability of a feature in a class

Learning in Bag of Words Naïve Bayes

- Parameters of the model:
 - Prior over the classes
 - Probability of a feature in a class
- For training, we assume N annotated examples and maximize their likelihood to find the maximum

$$L(\{x^{(i)}, y^{(i)}\}) = \prod_{i=1}^N P(x^{(i)}, y^{(i)}) = \prod_{i=1}^N P(y^{(i)}) \cdot P(x^{(i)}|y^{(i)}) = \prod_{i=1}^N P(y^{(i)}) \prod_{j=1}^d P(x_j^{(i)}|y^{(i)})$$

Learning in Bag of Words Naïve Bayes

- the Maximum Likelihood estimators turn out to be:

$$P(y = a) = \frac{|\{i : y^{(i)} = a\}|}{N}$$

$$P(x_j = 1|y = a) = \frac{|\{i : y^{(i)} = a \wedge x_j^{(i)} = 1\}|}{|\{i : y^{(i)} = a\}|}$$

- We need to apply some smoothing, obviously